

A Trip to Sockerbruket i Roma, 1954

Coding Assignment -- Questions for the Assistant

Kalle Gustavsson, a delegate from Stratsys Bageri AB in Stockholm, is travelling to Roma Sockerbruk on Gotland to negotiate a new weekly sugar supply contract. He has gathered a set of documents (price lists, timetables, correspondence, recipes, order records, and a guidebook) to prepare for the trip. Build an assistant that can answer the following questions using only the information in the provided documents.

Trip planning

1. Kalle needs to be at Roma Sockerbruk at 10:00 on Tuesday, 14 September 1954. Using the timetables provided, plan his full outbound journey from Stockholm: which train and boat departures should he take, and on what date must he leave Stockholm?
2. Plan Kalle's return journey. Which train and boat should he take back, and when will he arrive in Stockholm?
3. What is the total cost of Kalle's own travel (train and boat, round trip), assuming he travels in a 2-berth cabin on the ferry?

Pricing and the sugar deal

4. Based on the order book and the bakery's standard recipe, how many kilograms of sugar does Stratsys Bageri AB currently use per week?
5. Given that weekly volume, what is the best price per kilogram of sugar the bakery can obtain, assuming it signs a 12-month contract?
6. The bakery currently has its weekly sugar need shipped from Roma in one shipment per week. What would the total freight cost be per kilogram if it instead consolidated a full month's sugar into a single monthly shipment? Is this cheaper?
7. With a budget of 5,000 kr to spend on sugar, how many cinnamon buns could Stratsys Bageri AB produce? State any assumptions you make.

Sightseeing

8. Kalle has a free period in Roma from 13:00 to 16:00, after lunch and before his return train to Visby. If he wants to prioritize medieval history, what should he go and see?
9. After returning to Visby, Kalle has some free time before his evening departure. If he wants to see as much history as possible without arriving after anything has closed, what should he visit, and in what order?

Document reliability

10. An internal memo and the mill's official price list state different prices for sugar. Which price should be used when preparing for the negotiation, and

why?

Note: after reviewing your solution, we will also test it with a small number of additional questions of the same kind, using the same set of documents.